

1a. Algorithms and Problem Solving

****What is meant by the term algorithms in computation and outline the steps****

> ****Algorithm**** refers to the sequence of steps (methods) to be followed to solve a problem. Facilitate problem solving process in a computer and is the basis for coding a program.

****Steps involved in problem solving in computer:****

- * Define the problem
- * Analyze the problem (PADT2D)
- * Develop an algorithm to solve problems
- * Translate algorithm to computer language
- * Test and debug the program.
- * Document the program.

1b. Distinguish between Algorithms and Flowchart

Algorithm		Flow chart	
1) Representation of sequence of steps in english like language.		Diagrammatic representation of steps to be taken to solve a problem.	
2) A well defined set of instruction that takes inputs, processes it and produce output.		Illustrate sequence of steps, decision and connections between them.	
3) Can be expressed in various forms eg natural language, pseudocode.		Provide a clear and concise way to understand, communicate the algorithm or process.	

1c. Flowchart Symbols

Give flowchart symbols and the meaning as used in problem solving.**

> Answer:

> Flowchart Is a diagrammatic representation of steps to be taken to solve a problem. It uses various flow chart symbols connected by arrows.

- * start / stop
- * : Condition box
- * : arrows representing flow of instructions
- * : process
- * : Input / output operation
- * : stored data
- * : connector

1d. Programs & Classification

What is meant by the term program and identify its classification.

Answer: A set of instructions written in programming language, to convert raw data into desired useful information.

Classification:

1. High level: These are English-like and used by programmers to write programs. provides user friendly interface eg BLAST Web server.
2. Low-level: Comprise of machine and assembly level language, requires programming experience eg custom python scripts.

1e. Scientific Language

Scientific language refers to the specialized technology program used to develop scientific concepts, techniques, and research in the field. The language is often precise and technical allowing scientists to communicate complex data and data efficiency and accurately.

Examples Include:

- * FORTRAN (Formula Translation)
- * MATLAB
- * Python
- * Julia
- * C++
- * Java
- * ALGOL (Algorithmic language)
- * BASIC

1f. Introduction to Computational Chemistry

2q. Computational Chemistry

Computational chemistry is the branch of chemistry that uses computers, mathematical models, and simulations to study the structures, properties, and behavior of molecules and chemical reactions.

Reasons Why We Do Chemistry on Computer / CMS:

1. Safety: Invariably, all laboratory experiments carry some risk associated with chemicals or apparatus to be used. It is safer to perform a costly assessment prior to experimental work. Computational chemistry allows the user to carry out work on dangerous chemicals with no risk.
2. Cost: Apart from the financial outlay on a computer and associated peripherals together with the appropriate software, no further cost are involved unlike the experimental laboratory where most chemicals require disposal after use.
3. Understanding: Computational chemistry has the ability to provide a basis for understanding chemical principles.
- 4 Access to extremes

1g. Quantum Mechanics vs. Molecular Mechanics

Differences B/w Quantum Mechanics & Molecular Mechanics:

- | Features | Quantum Mechanics | Molecular Mechanics |
|--------------|------------------------------------|-------------------------------|
| 1. Principle | Based on the Schrodinger equation. | Based on classical mechanics. |

2. Treatment of particles	Considers electrons and nuclei.	Considers only atoms (nuclei).
3. Methods	Ab-initio, Semi-empirical	Empirical force fields
4. System size	Best for small molecules	Best for large molecules, DNA, proteins, polymers
5. Accuracy	Very high accuracy	Moderate accuracy

1h. Software Packages Used in Computational Chem

- * ChemDraw
- * HyperChem
- * AutoDock
- * Gaussian
- * Spartan
- * NWChem
- * MOPAC

1i. Theoretical vs. Computational Chemistry

Differences B/w Theoretical & Computational Chem:

Feature	Theoretical Chemistry	Computational Chemistry
1. Objective	To formulate and improve chemical theories.	To perform calculations and simulations based on theories.
2. Focus	Concepts, equations, and principles.	Numerical analysis, modelling, and predictions.
3. Computer Dependence	Not always dependent on computer.	Highly dependent on computer.
4. Nature of Work	Mostly conceptual and theoretical.	Mostly practical and computational.
5. Output	Theories, equations and models.	Numerical data, graphs and molecular structures.
6. Accuracy	Depends on assumptions and model.	Depends on both the theory and computational method used.

1j. The Schrödinger Equation

> Schrödinger equation is an eigen value equation:**

$$\hat{H} \Psi = E \Psi$$

> Where:

- > $\hat{H}$ is Hamiltonian Operator (describes the kinetic energy of the nuclei and electrons and the electrostatic interaction b/w individual particles).
- > E is the energy of the system.
- > Ψ is the wave function (describes the probability of finding an electron at particular set of co-ordinate).

>
>

1k. Terms in Computational Chemistry

1. Ab-Initio Method:

A computational chemistry method based entirely on fundamental quantum mechanical principles without using experimental data.

* Significance:

- * Provides highly accurate results
- * Predicts molecular structures and energies (E)

2. Semi-Empirical Method:

A quantum chemical method that simplifies calculations by incorporating some experimental data and approximations.

* Significance:

- * Faster than ab-initio methods (FR)
- * Requires less computational power

3. Empirical Method:

A method that relies mainly on experimental data and force fields rather than quantum mechanical calculations.

* Significance:

- * Very fast and computationally efficient.
- * Applied in molecular dynamics and drug design (VA).

4. Structure-Property Relationship (SPR):

The relationship b/w a molecule's chemical structure and its physical or chemical properties (e.g. boiling point, solubility, toxicity).

* Significance:

- * Predict properties of new compounds.
- * Help design materials with desired characteristics (pH).

5. Structure-Activity Relationship:

The relationship b/w a molecule's chemical structure and its biological activity.

* Significance:

- * Essential in drug discovery and development.
- * Assists in designing safer and more effective drugs (EA).

Statistical Mechanics

Explanation

Statistical mechanics relates the microscopic behavior of atoms and molecules to the macroscopic properties of matter.

Significance

- Explains temperature.
- Calculates entropy.
- Predicts pressure.
- Studies equilibrium.
- Connects molecular behavior with thermodynamics.

□

(f) Thermodynamics

Explanation

The branch of science dealing with heat, work, energy, and spontaneity of chemical reactions.

Important quantities:

- Enthalpy (H)
- Entropy (S)
- Gibbs free energy (G)

Significance

- Predicts reaction feasibility.
- Calculates equilibrium constants.
- Determines reaction energy.
- Studies phase changes.
- Optimizes industrial processes.

□

(g) Structural Properties

Explanation

These are physical characteristics that describe the arrangement of atoms in molecules.

Examples:

- Bond length
- Bond angle

- Molecular geometry
- Dipole moment
- Vibrational frequencies

Significance

- Determines molecular stability.
- Predicts reactivity.
- Explains physical properties.
- Assists drug design.
- Supports spectroscopic analysis.

□

(h) Quantitative Structure–Activity Relationship (QSAR)

Explanation

QSAR is a computational method that relates the chemical structure of compounds to their biological or chemical activity using mathematical models.

Significance

- Drug discovery.
- Predicts toxicity.
- Reduces laboratory experiments.
- Designs better pharmaceuticals.
- Saves cost and time.

□

(i) Modelling the Solid State

Explanation

The computational study of crystals and solid materials to predict their structures and properties.

Significance

- Designs new materials.
- Studies semiconductors.
- Investigates catalysts.
- Predicts crystal structures.
- Improves battery and electronic materials.

□

(j) Symbolic Calculations

Explanation

Symbolic calculations involve manipulating mathematical expressions exactly using

computer algebra software instead of numerical approximations.

Examples:

- Mathematica
- Maple

Significance

- Solves equations exactly.
- Simplifies complex formulas.
- Derives mathematical models.
- Reduces calculation errors.
- Assists theoretical chemistry.

□

(k) Artificial Intelligence (AI)

Explanation

AI uses machine learning and advanced algorithms to analyze chemical data, predict molecular properties, and accelerate scientific research.

Significance

- Accelerates drug discovery.
- Predicts molecular properties.
- Optimizes chemical reactions.
- Analyzes large datasets.
- Designs new materials and compounds

2. Spreadsheet Package (Excel Basics)

2a. Introduction to Excel

Spreadsheet package or excel** is an electronic spreadsheet application that enables user to store, reorganize, calculate, and manipulate the data with formulas using the spreadsheet system broken up by rows and column.

2b. Data Format in Excel

- * Number
- * Currency
- * Accounting
- * Date
- * Percentage
- * Text
- * Special
- * Custom
- * General

2c. Operational Parameters

Factors that Determine the choice of Spreadsheets:

- * Hardware requirement
- * Tech support
- * Output Quality
- * Documentation
- * Ease of learning/use
- * Price of spreadsheet package

Uses of Excel:

- * Track daily sales
- * Compare student result
- * Perform graphical analysis of data
- * Used in preparation of budgets
- * Compute profits and losses

2d. Excel

Advantages:

- * They are very easy to change
- * It makes data organization more efficient
- * It offers third party support
- * It makes it easy to create instant calculations
- * They are integration options

Disadvantages:

- * It takes time to learn how to use formulas
- * Errors may not be easily detected
- * It takes time to enter data into cells
- * Spreadsheet are expensive

Features of Excel:

- * cell
- * cell name
- * cell reference
- * Active cell
- * Column
- * chart
- * Formula bar
- * Name box
- * Rows
- * column

3. Functions in Excel

3a. Definitions of Core Functions

1. MAX: Maximum in excel will return the largest value from a given range. ``Max(Val1, Val2)``
2. MIN: Minimum will return the smallest value from a given range. ``Min(Val1, Val2)``
3. MOD: Will return remainder. ``Mod(no, divisor)``
4. Round: In excel rounds the given number to a specific number of digits. ``ROUND(no, no of digits)``
5. Round down: Round a given number towards zero. ``Round down (no, no of digits)``
6. Sum: In excel adds 2 given numbers or a given range of cells.
7. EVEN: will round up to nearest even integral or number. ``Even(no)``
8. Odd: will round up to nearest odd number. ``Odd(no)``
9. Average: Average in Excel returns the average of the argument in specified range. ``Average(no1, no2)``
10. Count: In excel counts the no of cell in the specified range.
11. Count blank: This returns all blank cells within the specified range.
12. Large value In excel returns the nth largest value for a given specified range.

3b. Excel Function Practice Examples

Question:

Assuming A₁=10, A₂=15, A₃=20, A₄=25

Get (i) Count, (ii) Count blank if (A2 to A4 is empty), (iii) Large value, (iv) Minimum, (v) Average, (vi) Mode

Solutions worked out in notes:

i) AVE (A1:A4, 7):

A₁=10, A₂=15, A₃=20, A₄=25

$$\text{AVE}(A1:A4, 7) = \frac{10+15+20+25+7}{5} = 15.4$$

COUNT (A1:A4\$

= 4

Count A1:A4, 7

= 5

Countblank (A1:A4, if A2 is empty):**

COUNTBLANK}(A1:A4) = 0

Countblank = 3

MAX A1:A4

= 25

MIN A1:A4

= 10

MIN (A1:A4, 7

= 7

MOD Calculations:

{Mod}(A1, A2) = 10 divide 15 = 0 r 10 = 10

Mod}(A2, A1) = 15 divide 10 = 1 r 5 = 5

Mod A1, 3) = 10 divide 3} = 3 r 1 = 1

Mod}(A3, 6) = 20 divide 6} = 3 r 2 = 2

#4. Excel Rounding, Tables, & Charts

4a. Round Up and Round Down Exercises

Assuming the cell A1 contains or holds the no 2365.854. Using get roundup and rounddown in the following ways:

1. Round up}(A1, 0) = 2366
2. Round up}(A1, 1) = 2365.9
3. Round up}(A1, 2) = 2365.86
4. Round up}(A1, 3) = 2365.854
5. Round up}(A1, -2) = 2400

Rounddown A1

1. Rounddown}(A1, 0) = 2365
2. Rounddown}(A1, 1) = 2365.8
3. Rounddown}(A1, 2) = 2365.85
4. Rounddown}(A1, 3) = 2365.854
5. Rounddown}(A1, -2) = 2300

4b. Excel Tables

An Excel table is a structured range of data that is organized into rows and columns, allowing easy sorting, filtering and analysis of information.

Characteristics:

- * Fast filtering, switching, and sorting are easy.
- * It allows formatting with In-built table styles.
- * Column headings are always visible when scrolling.
- * Formulas are easy to read using specialized syntax.

4c. Excel Charts

They are used in excel to represent data pictorially.

Types:

- * Column chart: used to compare values across categories.
- * Bar chart: Useful when category name is long.
- * Line chart: Shows trends over a time of continuous data, useful for reaction rate analysis.
- * Pie chart: shows percentage of proportion.
- * Scattered plot or x - y chart: are important in chemistry used for relationship b/w various variables.
- * Area chart: used for magnitude of change over time.
- * Doughnut chart

4d. Common Errors in Chart Creation

1. Selection of Incorrect data range.
2. Using Inappropriate chart type.
3. Missing Axis label.
4. Poor Chart title.
5. Overcrowded data presentation.
6. Incorrect scaling of axis.

4e. Advantage & Limitation of Chart

Advantage	Limitation
① It is easy to use	It requires computer literacy.
② Fast data analysis	Error may occur with wrong data Input.
③ Used for professional presentation	Complex chart may confuse beginners.
④ Supports scientific calculation	Depends on software availability.

4f. Steps to Create a Chart

1. Enter data into rows and columns.

2. Highlight the data range.
3. Click the Insert tab.
4. Choose appropriate chart type.
5. Customize chart using chart design tools.
6. Add chart title, axis label and legend.
7. Save and print chart if necessary.

Here is the comprehensive transcription of the handwritten notes from your images, organized logically into Excel topics and Computational/Quantum Chemistry topics.

1. Excel Tables and Charts

> Excel table work as a whole, its helps you manage a table content independently from the sheet data.

Characteristics

1. They are dynamic by nature and very easy to add and remove columns and rows.
 2. Filtering and sorting are easily integrated.
 3. It allows formatting with inbuilt table styles.
 4. Columns heading are always visible while scrolling.
 5. Quick total make sum, count data, minimize, maximize and others.
 6. Using calculated columns helps to compute entire column in a cell but void of formular.
 7. Formulas are easy to read due to special syntax.
 8. Dynamics charts cannot be adjusted while adding and removing data in a table.
- Note: Table use charts and Pivot table to prepare data.

Note: Table has the following three options:**

- > 1. You can sort and filter in a table.
- > 2. You can format.
- > 3. You can do auto filling.

How do we create a table?

1. Select any cell within your data cell.
2. On the Insert tab in the table group, click the **table** button.

Advantages

1. Easy to create
2. Easy to read
3. They are auto-fill
4. They change automatically

5. They updated automatically.

Practice Example

Question: Describe a data range in which the first cell is \$A_1\$ and the last cell is \$G_8\$ in the employing salary spreadsheet.

	A	B	C	D	E	F	G
1	Employee Number	Last Name	First Name	Dept.	Type of Job	Salary	Bonuses
2							
3							
4							
5							
6							
7							
8							

2. Spreadsheet Charts Deep Dive

> Charts in Excel are used to represent data pictorially.

Core Concepts

- * Data Series: It's a group of related data points to display chart.
- * Category Axis: It's usually the horizontal axis .
- * Value Axis: It's usually the vertical axis .
- * Legend: This explains chart symbols and columns.
- * Grid lines: These are lines that increase readability.
- * Plot area: They are used to display regions where data are displayed.

Types of Charts

1. Column chart: Used to compare value across categories.
2. Bar chart: They are used when categories names are long.
3. Line chart: This shows trends over a time of continuous data, they are used for reaction rate analysis.
4. Pie-chart: This shows percentage of proportions.
5. Scattered chart or X and Y chart: (Are important in chemistry), it is used for relationship between variables.
6. Area chart: It is used for magnitude of change over time.
7. Histogram: Frequency distribution of data.
8. Radar chart: Compare multiple variables simultaneously.
9. Cylinder chart

Steps used for creating chart in Excel

1. Enter data into rows and column.

2. Highlight the data range.
3. Click the **Insert** tab.
4. Choose the appropriate chart style.
5. Customize the chart using chart design tools.
6. Add chart title, axis labels and legend.
7. Save and print the chart if necessary.

> What is Trendlines?

This is used to identify patterns and relationships in data.

Common Errors in Chart Creation

1. Selecting incorrect data range.
2. Using inappropriate chart style.
3. Missing axis labels.
4. Poor chart title.
5. Overcrowded data presentation.
6. Incorrect scaling of axis.

Advantages and Limitations of Charts

> Microsoft charts are powerful tools for scientific data analysis and presentation. You must know how to create and interpret charts in your lab experiment, work or research activities.

#3. Practical Assignment: Acid-Base Titration Graph

Context: During an Acid-base titration experiment, pH values are recorded after adding acid or base. A line chart is used to show the relationship between volume added and pH.

Data Table

Volumes of NaOH mL		
---	---	
0	2.0	
5	2.5	
10	3.1	
15	4.0	
25	7.0	
30	10.5	

Procedure

1. Enter the data.
2. Highlight the data.
3. Click Insert → line chart.

4. Add chart title and axis label.
5. Interpret the curve to determine the equivalent point.

Required Graphs

1. Titration curve $\rightarrow$ line graph (Acid-base titration).
2. Equivalency point.

4. Computational and Quantum Chemistry Basics

Computational Chemistry is the branch of chemistry that uses computer simulations and mathematical methods to study chemical structures, properties and reactions. It is an offshoot of physical chemistry that developed as a result of algorithms.

Reasons Why We Do Chemistry on a Computer

1. To save time and reduce experimental cost.
2. To predict molecular properties before experiments.
3. To study systems that are corrosive, harsh, or dangerous.
4. It is an alternative to wet chemistry and an improvement to instrumental chemistry.

Quantum vs. Molecular Models

Quantum	Molecular
i) It has a definite path and radius.	It does not have a definite path; it is a probability region.
ii) It is represented by shell names.	It is represented by s, p, d, f orbitals.

Classical Chemistry vs. Modern Chemistry

Classical Chemistry	Modern Chemistry
1) Based mainly on observations and experimental results.	Based on advanced theories and modern technology.
2) Does not explain atomic and molecular behaviour in detail.	Explains atomic and molecular behaviour using quantum mechanics and advanced models.
Example:Qualitative analysis	Example: Computational chemistry

Distinguishing Quantum Mechanics and Molecular Mechanics

Quantum Mechanics	Molecular Mechanics
1) It is a branch of science that studies the behaviour of matter and energy at the atomic and subatomic levels (proton, atom).	It is a computational method that uses the principles of classical mechanics to study the structure and properties of molecules.

5 Core Software Packages

1. Gaussian
2. Gromacs
3. Chemaxon
4. Hyperchem
5. Quantum Espresso

what are scientific languages

are specifically designed for solving scientific, engineering and mathematical problems. they are used for numerical calculations, simulations and data analysis
examples are: FORTRAN

MATLAB

PYTHON

PASCAL

C